PAPER_346

■# PAPER<i>346 — M87 Jet Blandford-Znajek Model: Full F</i>U<i>Bi</i>i Calculation

Author: Daniel T. Murphy

Session: 96

■# PAPER*346 — M87 Jet Blandford-Znajek Model: Full F*UB*i*i Calculation

Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gok*share*31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF F*U*Bi_i calculation for M87 with Blandford-Znajek power coupling **Author:** Daniel T. Murphy

<!-- UQFF constants: κ = 5.0e-4 day^{-1}, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

The complete UQFF buoyancy-unified force F*U*Bi*i is computed for M87* (Virgo A), the first black hole directly imaged by the Event Horizon Telescope. The Blandford-Znajek (BZ) jet power* PBZ = $B^2 r g^2$*c is connected to the UQFF* ωact = 2π/day rotational activation frequency, yielding F*U*Bi*i ≈ −8.32×10^{21}■ N. The X-ray jet luminosity L*X ≈ 10■^{0} W sets the luminosity scale for UQFF-M87 calibration.

---

2. Core Physics

2.1 UQFF Buoyancy-Unified Force

```
F_{U\_Bi\_i} = \frac{U_g e^{+/-}}{r^2} + F_{\rm Bi} + F_{\rm U} + F_{\rm react}
```

Full 5-component UQFF form yields:

```
F_{U\_Bi\_i} \approx -8.32 \times 10^{217}\ \mathrm{N}
```

2.2 Blandford-Znajek Jet Power

```
P_{\rm BZ} = \frac{\kappa_{\rm BZ}}{4\pi c} \Phi_{\rm BH}^2 \Omega_{\rm H}^2
f(\Omega_{\rm H})
```

Simplified as:

```
P_{\rm BZ} = B^2 r_g^2 c
```

where r*g* = *GM*BH/c^2 is the gravitational radius for M87* (M_BH = 6.5×10■ M■).

2.3 UQFF Activation Frequency

```
\omega_{\rm act} = \frac{2\pi}{T_{\rm day}} = \frac{2\pi}{86400\ \mathrm{s}} = 7.27
\times 10^{-5}\ \mathrm{rad/s}
```

The "one day" period corresponds to the characteristic M87 jet variability timescale observed in Very Long Baseline Array (VLBA) monitoring.

2.4 Gravitational Radius

```
r_g = \frac{G M_{\rm BH}}{c^2} = \frac{6.674\times 10^{-11} \times 6.5\times 10^9 \times
1.989\times 10^{30}}{(3\times 10^8)^2} \approx 9.6 \times 10^{12}\ \mathrm{m}
```

---

3. Key Values

Quantity	Formula	Value
M_BH	EHT 2019	6.5×10■ M■
F*U*Bi_i	UQFF full 5-eq	-8.32×10^{21}■ N
P_BZ	$B^2 r\_g^2 c$	~10■■ erg/s
ω_act	2π/day	7.27×10⁻■ rad/s
L_X	Chandra observation	~10■^{0} W
r_g	GM/c^2	9.6×10^{12} m

---

4. Physical Significance

M87 is the prototype for supermassive black hole jet physics. The UQFF F*U*Bi*i ≈ –8.32×10^{21}■ N is the baseline force scale for AGN-class black holes with M*BH ~ 10■ M■. The Blandford-Znajek coupling connects UQFF to the standard magnetically-arrested disk (MAD) jet framework, providing a bridge between UQFF vacuum buoyancy and electromagnetic jet extraction. The ω_act = 2π/day activation frequency is consistent with M87*'s Variable Emission Region (VER) coherence timescale.

---

5. Deduplication Note

vs. PAPER_347 (Centaurus A): M87 uses ω_act = 2π/day; Centaurus A uses ω_act = 2π/(12.5 yr). The BZ power forms are similar but calibrated to different AGN jet morphologies.

vs. PAPER_349 (SPT-CL J2215): SPT-CL J2215 yields the HIGHEST F_U_Bi_i in the PAPER_346–352 series at –1.40×10^{21}■ N.

---

6. Classification

Physics Territory: FIRST UQFF F*U*Bi_i for M87* with BZ jet power coupling **Scale:** Galactic (6.5×10■ M■ AGN) **CP Implementation:** `M87JetBZModelFUBiCalculator` (CondensedPhysics3.py, Session 96)
